MBC-DG-LAB Protocol

MBC-DG-LAB uses an USB to RS232 converter to communicate. Below, the list of commands:

Communication Channel:

RS232 link settings:

115200bps

1 stop bit

No parity

No data flow control

Set command:

Allows you to change parameters

Commands	Parameters	Possible values (#)	Step	Unit	Examples
AC ####	Auto Calibration	QUAD/MIN/MAX	NA	NA	AC QUAD
RM ####	Running mode	AUTO or MAN	NA	NA	RM AUTO
TL #####	Transfer level	-1000 to 1000	1	%*10	*TL –233*
DA ####	Dither amplitude	1 to 1000	1	mV	*DA 110*
DF ####	Dither frequency	400 to 1400	40	Hz	*DF 720*
PG #	Photodiode gain	1 to 127	1		*PG 3*
PP ####	Photodiode polarity	INV or NOT			*PP INV*
BV #####	BIAS value	-10000 to 10000	1	mV	*BV +4521*
SR	Scan Restart	NA	NA	NA	*SR*
SC	send all MBC parameters	NA	NA	NA	*SC*
SV #	Save MBC parameters to one of the config.	1/2/3	NA	NA	*SV 1*
LD #	Load MBC parameters to one of the config.	1/2/3	NA	NA	*LD 3*

Each command ends by a carriage return (carriage return ‘CR’ code ASCII 13)

If the command is valid, the system returns the sent value or « OK »

If an error happens, the system returns: « ERR »

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GET commands:

Returns a parameter's value

Commands	Parameters	Possible returned values	Units
RM?	Running mode	AUTO or MAN	NA
TL?	Transfer level	-1000 to 1000	%
DA?	Dither amplitude	1 to 1000	mV
DF?	Dither frequency	400 to 1400	Hz
PG?	Photodiode gain	1 to 127	-
PP?	Photodiode polarity	INV or NOT	
BV?	BIAS value	-10000 to 10000	mV
SN	Card serial number	000001	
F	Firmware version	V2.1	

Each command ends by a carriage return (carriage return 'CR' code ASCII 13)

If the command is valid, the system returns the value of the corresponding parameter.

If an error happens, the system returns: « ERR »

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